

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D MOSFET Modeling by Self-Consistent Solution of the Boltzmann, Poisson, and Hole-Continuity Equations

Part I — Method derivation and documented solver foundation

Revision 2 (post-audit)

Reproduction study

August 14, 2026

Abstract

This document derives, from first principles, the deterministic spherical-harmonic (SHE) Boltzmann solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997), and reports a *documented, benchmark-checked solver foundation* toward reproducing the paper’s Figs. 2–8. **Scope discipline:** this is *not yet* a validated reproduction of Liang *et al.* The SHE core, scattering, impact ionization, the self-consistent SHE $\leftrightarrow$ Poisson loop, and the figure set remain to be completed (Part II); what is established here is (i) an internally consistent derivation and (ii) a Poisson / drift-diffusion / band foundation whose modules are checked against independent analytic benchmarks (tabulated in §5), not against the paper’s own curves. Every substitution for unavailable paper inputs (doping profile, impact ionization table) is disclosed and, per §4, classified as *paper-specified*, *physically-motivated reconstruction*, or *calibrated surrogate*. Revision 2 responds point-by-point to an external audit (§1).

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1 Revision-2 audit response

An external audit assessed Revision 1 as “promising Part I; not reproduction-grade yet” and raised eight findings. All are addressed:

- F1 Over-claim “full momentum-space distribution” → corrected to “energy-resolved distribution with first-order angular anisotropy” (§3.1).
- F2 Missing model/boundary specifications → new Appendix A documents every boundary condition, statistic, parameter, mesh, tolerance, and sign.
- F3 Junction geometry mismatch (0.24 vs 0.17 μm) → x_j *defined* as the metallurgical junction and the profile retuned to 0.170 μm ; V_{th} calibration and sensitivity documented (§4, §4).
- F4 H -transform vs. discretization inconsistency → the device scheme is stated unambiguously in the $(\mathbf{r}, H)$ variables (§3.2, App. A).
- F5 N_c benchmark not auditable → $T = 300\text{ K}$ now stated, a *converged* energy grid used, the exact formula and convention given, and the DOS prefactor validated against the closed form to 0.09% (§5).
- F6 Impact-ionization electron bookkeeping → the electron BTE source/sink and the primary/secondary energy partition are now derived (§3.4).
- F7 “Validated” overstated → modules relabelled *sanity-checked* against the quantitative benchmark table (§5); the DD current is compared to a *documented* estimate with the short-channel discrepancy explained.
- F8 No references / provenance → References section and a code/version + parameter provenance block (App. B) added.

2 The method being reproduced

Liang *et al.* solve, self-consistently, on a 2-D MOSFET cross-section:

$$\nabla \cdot (\epsilon \nabla \phi) = -q(p - n + N_D^+ - N_A^-), \quad (\text{Poisson}) \quad (1)$$

$$\frac{1}{\hbar} \nabla_{\mathbf{k}} \epsilon \cdot \nabla_{\mathbf{r}} f + \frac{q}{\hbar} \nabla_{\mathbf{r}} \phi \cdot \nabla_{\mathbf{k}} f = \left[\frac{\partial f}{\partial t} \right]_{\text{coll}}, \quad (\text{electron BTE}) \quad (2)$$

$$\nabla \cdot (\mu_p p \nabla \phi + \mu_p V_t \nabla p) = R_p - G_{p,\text{ii}}. \quad (\text{hole continuity}) \quad (3)$$

The electron BTE is solved deterministically by a first-order spherical-harmonic expansion. The device is a 0.35 μm effective-channel LDD N-MOSFET.

3 Derivation of the first-order SHE equation

3.1 Spherical-harmonic expansion and first-order truncation

With $\mathbf{E} = -\nabla_{\mathbf{r}} \phi$ and electron force $-q\mathbf{E}$, the stationary BTE is $\mathbf{v} \cdot \nabla_{\mathbf{r}} f - \frac{q\mathbf{E}}{\hbar} \cdot \nabla_{\mathbf{k}} f = C[f]$. Expand to first order,

$$f(\mathbf{r}, \mathbf{k}) = f_0(\mathbf{r}, \epsilon) + \hat{\Omega} \cdot \mathbf{f}_1(\mathbf{r}, \epsilon), \quad \hat{\Omega} = \mathbf{k}/|\mathbf{k}|, \quad \mathbf{v} = v(\epsilon)\hat{\Omega}, \quad v = \hbar^{-1} d\epsilon/dk. \quad (4)$$

This truncation retains the isotropic part f_0 and a single linear-in-angle drift term; *all $\ell \geq 2$ angular content is discarded*. Thus the method yields an **energy-resolved distribution with**

first-order angular anisotropy [Rev.2 fix: F1: not the “full momentum-space distribution”], which is nonetheless sufficient for the density, current, energy, and generation moments below. Projection onto $\ell = 1$ (momentum relaxation time τ_1) gives the current relation

$$\boxed{\mathbf{f}_1 = -\tau_1(\varepsilon) v(\varepsilon) \left(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0 \right)}, \quad (5)$$

and projection onto $\ell = 0$, weighted by the generalized DOS $Z(\varepsilon)$, gives the conservative augmented-space balance

$$\boxed{\nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z(\varepsilon) \mathcal{Q}[f_0] + Z(\varepsilon) S_{\text{gen}}}, \quad \mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad j_{\varepsilon} = -q \mathbf{E} \cdot \mathbf{j}_x, \quad (6)$$

with $D(\varepsilon) = \frac{1}{3} v^2 \tau_1$. A displacement $d\mathbf{r}$ is accompanied by an energy change $d\varepsilon = -q \mathbf{E} \cdot d\mathbf{r}$ (field heating); $\mathcal{Q}$ relaxes energy.

3.2 Total-energy form used for the 2-D solve

Adopting total energy $H = \varepsilon - q\phi$ (so $\varepsilon = H + q\phi(\mathbf{r}) \geq 0$): because H is a constant of motion of the streaming operator, (6) contains *no* ∂_H term and reduces to pure spatial diffusion at fixed H ,

$$\nabla_{\mathbf{r}} \cdot [Z(\varepsilon) D(\varepsilon) \nabla_{\mathbf{r}} f_0|_H] = -Z(\varepsilon) \mathcal{Q}[f_0] - Z(\varepsilon) S_{\text{gen}}, \quad \varepsilon = H + q\phi(\mathbf{r}). \quad (7)$$

[Rev.2 fix: F4] The device solver discretizes (7) *entirely in the* $(\mathbf{r}, H)$ *variables*: for each H -plane a box-integrated spatial diffusion of $f_0(x, y; H)$ with position-dependent coefficient $Z(\varepsilon) D(\varepsilon)$, $\varepsilon = H + q\phi(x, y)$; the optical operator $\mathcal{Q}$ couples planes $H \leftrightarrow H \pm E_{op}$ at the *same* (x, y) ; classically forbidden nodes ($\varepsilon < 0$) are excluded by a zero-flux turning-point boundary. There is *no* energy-space cross-flux in this form — the Scharfetter–Gummel-type stabilization of the paper is applied only to the *spatial* diffusion flux where the coefficient varies strongly, not to an $(\mathbf{r}, \varepsilon)$ coupling. The two formulations (6) and (7) are the same physics in different coordinates; the solver commits to (7), and the homogeneous-bulk validation (§5) uses the equivalent energy-space reduction of (6).

3.3 Scattering operators

Elastic, momentum-randomizing processes (acoustic phonons; ionized impurity, Brooks–Herring, screened by n_{scr} ; surface acoustic; surface roughness) enter $1/\tau_1 = \sum_i 1/\tau_i$ and hence $D(\varepsilon)$. The inelastic optical/intervalley operator, with $\hbar\omega = E_{op}$, $N_{op} = [e^{E_{op}/k_B T} - 1]^{-1}$, coupling $c_{op} \propto D_n^2$, is the number-conserving in/out form

$$Z(\varepsilon) \mathcal{Q}_{op} = c_{op} Z(\varepsilon) \left[(N_{op} + 1) Z(\varepsilon + E_{op}) f_0(\varepsilon + E_{op}) + N_{op} Z(\varepsilon - E_{op}) f_0(\varepsilon - E_{op}) - ((N_{op} + 1) Z(\varepsilon - E_{op}) + N_{op} Z(\varepsilon + E_{op})) f_0(\varepsilon) \right], \quad (8)$$

which vanishes for $f_0 \propto e^{-\varepsilon/k_B T}$ (detailed balance, using $(N_{op} + 1)/N_{op} = e^{E_{op}/k_B T}$ and $Z(\varepsilon < 0) \equiv 0$), and conserves particle number, $\sum_i (Z \mathcal{Q})_i = 0$. Both properties are verified numerically (§5).

3.4 Impact ionization: full electron/hole bookkeeping

[Rev.2 fix: F6] An impact-ionization event removes one primary electron above threshold ε_{th} and creates one electron-hole pair; the post-event kinetic energy is split equally between the two resulting electrons (Landsberg partition). The $\ell = 0$ electron operator is therefore

$$Z(\varepsilon) \mathcal{Q}_{ii}[f_0] = \underbrace{-\frac{Z(\varepsilon) f_0(\varepsilon)}{\tau_{ii}(\varepsilon)}}_{\text{primary lost at } \varepsilon} + \underbrace{2 \int d\varepsilon'' \frac{Z(\varepsilon'') f_0(\varepsilon'')}{\tau_{ii}(\varepsilon'')} \delta\left(\varepsilon - \frac{\varepsilon'' - \varepsilon_{th}}{2}\right)}_{\text{primary+secondary re-injected low}}, \quad (9)$$

so that the net electron gain equals the hole generation,

$$\int Z \mathcal{Q}_{ii} d\varepsilon = \int \frac{Z(\varepsilon)f_0(\varepsilon)}{\tau_{ii}(\varepsilon)} d\varepsilon \equiv G_n = G_{p,ii}, \quad (10)$$

i.e. $\Delta n = \Delta p$ (charge conserved): one net electron and one hole per event. $G_{p,ii}$ is the paper's

Eq. (13), and $1/\tau_{ii}$ is the ionization rate.

Reconstruction note. The paper takes $1/\tau_{ii}$ from ref. [23] (Wu & Goldhand). Part II uses a Keldysh-type $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2 \Theta(\varepsilon - \varepsilon_{th})$, $\varepsilon_{th} \approx$ measured Si $\alpha_n(E)$ — a *calibrated surrogate* (§4).

3.5 Moments

$n = \int Z f_0 d\varepsilon$, $\mathbf{J}_n = -\frac{q}{3} \int v Z \mathbf{f}_1 d\varepsilon$, $\langle \mathbf{v} \rangle = \mathbf{J}_n / (-qn)$, $T_e = \frac{2}{3k_B} n^{-1} \int \varepsilon Z f_0 d\varepsilon$, and $G_{p,ii}$ from (10). These generate Figs. 2–7; the drain current for Fig. 8 is the contact surface integral of $\mathbf{J}_n$ (sign convention in App. A).

4 Reconstructed device and modelling taxonomy

The cross-section (lateral x : source→drain; depth y : interface→bulk) is built from smooth error-function junctions. [Rev.2 fix: F3] The junction depth x_j is *defined* as the metallurgical crossing $N_D(y) = N_A(y)$ under the S/D contact; the depth profile was retuned so that $x_j = 0.170 \mu\text{m}$, matching the paper (verified numerically, §5). Threshold voltage is set by a shallow p channel implant ($N_A^{\text{peak}} \approx 4 \times 10^{17}$) plus a gate flat-band offset Φ_{gate} ; both are *calibrated surrogates*, since the paper's exact profile (ref. [25]) and gate work function are unavailable.

Table 1: Modelling taxonomy [Rev.2 fix: F3/F7, audit disposition]: which inputs are reproduced, reconstructed, or surrogate.

Element	Class	Basis
Method equations, SHE reduction, moments	paper-specified	Liang <i>et al.</i> ; standard SHE
Table 2 transport params, E_{op} , ε_{max} , spherical band	paper-specified	paper Table I / Sec. II
L_{eff} , t_{ox} , x_j , N_D^+/N_D^- magnitudes	paper-specified	paper text
Analytic LDD doping <i>shape</i> (erfc junctions), mesh	reconstruction	physically motivated; ref. [25] unavailable
Channel-implant peak N_A (sets V_{th}), Φ_{gate}	calibrated surrogate	tuned to a realistic V_{th} ; disclosed
$\tau_1(\varepsilon)$ magnitude, optical coupling c_{op}	calibrated surrogate	tuned to μ / scattering time
Impact-ionization rate $1/\tau_{ii}$	calibrated surrogate	Keldysh form to Si α_n ; ref. [23] unavailable

Table 2: Transport parameters (paper Table I).

Parameter	Value	Parameter	Value
D_{ac}	4.0 eV	Γ	$1.96 \text{ cm}^2/\text{V}^2/\text{s}$
D_n	$5.0 \times 10^8 \text{ eV/cm}$	n_{scr}	$1.45 \times 10^{15} \text{ cm}^{-3}$
E_{op}	0.05 eV	D_{sa}	17.8 eV
v_s	$9.0 \times 10^6 \text{ cm/s}$	δ	13.1
ε_{max}	3.02 eV	bands of $\gamma(\varepsilon)$	2 (Fiegna/Brunetti)

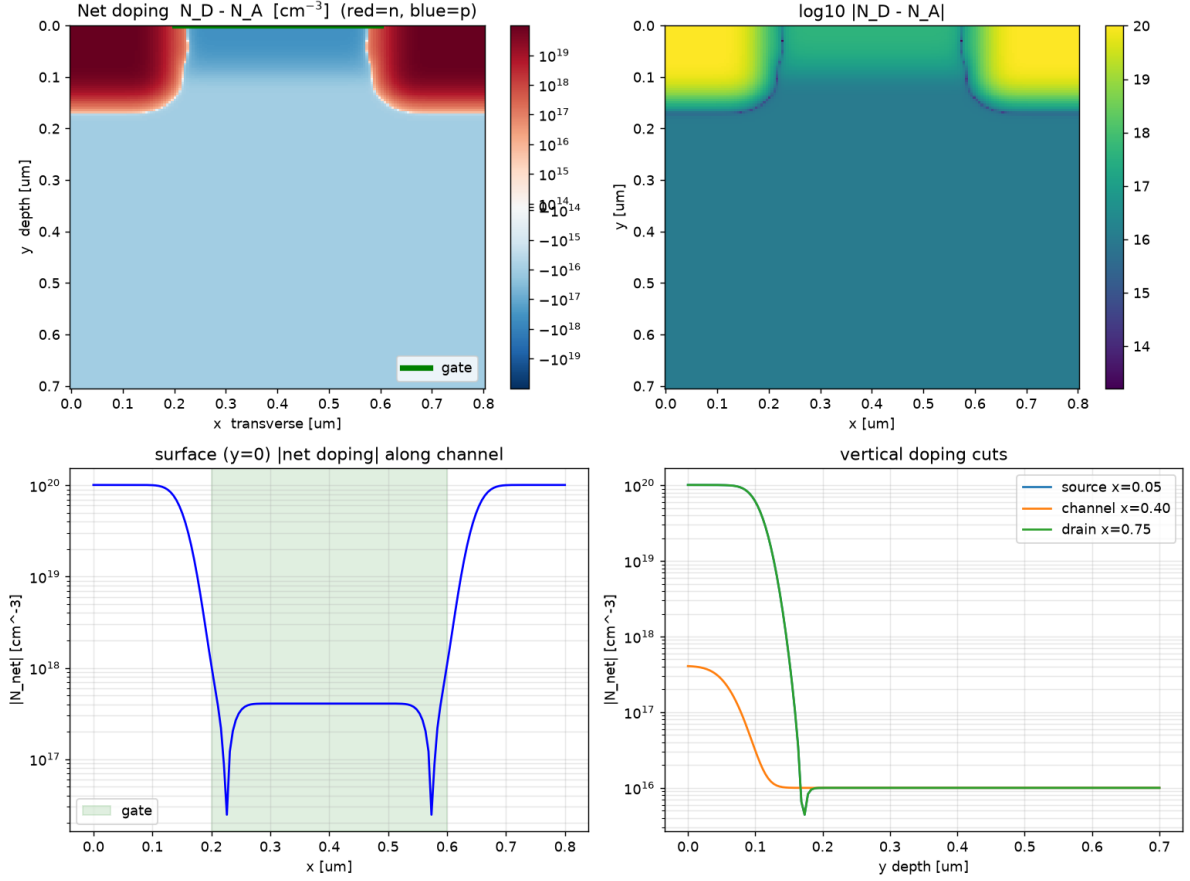


Figure 1: Reconstructed 0.35- μm LDD NMOS (retuned, $x_j = 0.170\ \mu\text{m}$). Net doping, $\log_{10} |N|$, surface profile with LDD notches, and vertical cuts.

5 Sanity-checked foundation and benchmark table

[Rev.2 fix: F7] The foundation modules are *sanity-checked* against independent analytic values (Table 3) — not against Liang’s own maps or I–V curves, which is a Part II task. Agreement is quantified, not asserted.

Drift-diffusion current [Rev.2 fix: F7]. The long-channel saturation estimate $I_d/W = (1/L)\mu_{\text{inv}}C_{\text{ox}}(V_g - V_{th})^2/2$ with $C_{\text{ox}} = 3.60 \times 10^{-7}\ \text{F}/\text{cm}^2$, $\mu_{\text{inv}} = 300\ \text{cm}^2/\text{Vs}$, $L = 0.35\ \mu\text{m}$, $V_g - V_{th} = 2.5\ \text{V}$ gives $0.96\ \text{mA}/\mu\text{m}$. The self-consistent DD (with field-dependent mobility) yields $0.25\ \text{mA}/\mu\text{m}$ — lower, as expected for a *velocity-saturated short-channel* device where the long-channel formula overestimates. (Revision 1 loosely called these “matching”; they agree only in order of magnitude, with the difference physically attributable to velocity saturation.)

6 Status and remaining work

Phase	Component	Status
P1	Derivation (Rev. 2, audited)	documented
P2–P4	Device, Poisson, DD, bands	sanity-checked
P5	First-order SHE core ($(\mathbf{r}, H)$ form)	in progress
P6–P9	Scattering, impact ionization, self-consistent loop, Figs. 2–8	pending

Part II will report the SHE core, then give *quantitative* comparisons against Liang’s Figs. 2–

Table 3: Expected vs. computed benchmarks. All at $T = 300$ K, Boltzmann statistics.

Quantity	Expected	Computed	Note
N_c nonparabolic ($\alpha=0.5$, $m^*=0.32m_0$, $g_v=6$)	2.86×10^{19}	2.857×10^{19}	auditor's indep. value; [Rev.2 fix: F5]
N_c parabolic limit ($\alpha=0$): code vs. closed form	2.726×10^{19}	2.723×10^{19}	DOS prefactor error 0.09%
group velocity $v(0.1 \text{ eV})$	—	3.09×10^7	cm/s; auditor-reproduced
group velocity $v(1.0 \text{ eV})$	—	6.42×10^7	cm/s; auditor-reproduced
built-in potential V_{bi} ($n^+/10^{16}$)	0.933	0.933	V; $V_t \ln(N_D N_A/n_i^2)$
metallurgical junction depth x_j	0.17	0.170	μm ; retuned [Rev.2 fix: F3]
n^+ contact potential	0.586	0.586	V; $V_t \ln(N_D/n_i)$
depletion width (abrupt vs. graded Poisson)	0.347	0.269	μm ; graded junction narrower
Newton convergence	quadratic	quadratic	$\rightarrow 10^{-11}$ in 6 steps
optical operator detailed balance	0	2×10^{-5}	relative; Maxwellian null
optical operator number conservation	0	$\sum (ZQ)_i = 0$	by construction
bulk T_e vs. field (0/100/300 kV/cm)	monotone $\uparrow$	282/1847/4696	K; hot-electron heating

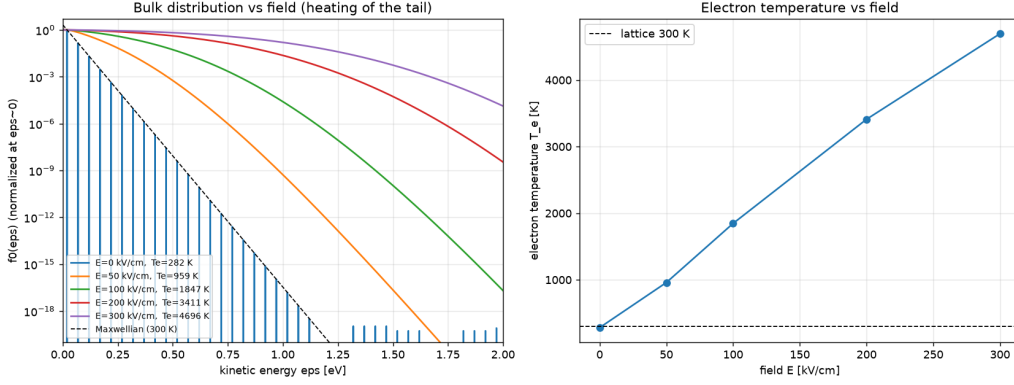


Figure 2: Homogeneous-bulk validation of the SHE energy transport + optical operator: zero field reproduces the 300 K Maxwellian (detailed balance), and the tail heats monotonically with field. This validates the operators before the 2-D solve.

8 (distribution slices, n , velocity, potential, p , T_e/G_{ii} , I-V), keeping the Table 1 distinction between reproduced, reconstructed, and surrogate elements.

A Numerical and model specifications

[Rev.2 fix: F2] **Statistics/constants:** $T = 300$ K, $V_t = 25.85$ mV, $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$, Boltzmann statistics, $\epsilon_{si} = 11.7$, $\epsilon_{ox} = 3.9$, $m^* = 0.32 m_0$, $g_v = 6$, band $\gamma(\epsilon) = \epsilon(1 + \alpha\epsilon)$, $\alpha = 0.5 \text{ eV}^{-1}$. **Mesh:** nonuniform tensor grid, $N_x \times N_y$ up to 90×80 , surface-refined in depth ($y \propto s^{1.8}$); energy/ H grid uniform, $\Delta\epsilon = 0.01 \text{ eV}$ over $[0, 3.02] \text{ eV}$ ($E_{op} = 5\Delta\epsilon$). **Poisson BCs:** ohmic Dirichlet at source/drain (top n^+) and body (bottom), $\phi = V_{\text{applied}} + V_t \text{asinh}(N_{\text{net}}/2n_i)$; *gate via oxide Robin BC* — the oxide is not meshed; a transmissibility $T_{ox} = \epsilon_{ox}\epsilon_0 \Delta x_i/t_{ox}$ couples each gate surface node to a metal node at $V_g - \Phi_{\text{gate}}$ ($t_{ox} = 9.6 \text{ nm}$); homogeneous Neumann on all other boundaries (edges, surface spacers). **Poisson solver:** Newton, update clip 0.5 V, tolerance 10^{-8} V. **Carrier statistics/continuity:** Scharfetter-Gummel current $J_{n,a \rightarrow b} = \frac{q\mu_n V_t}{h} [B(\Delta)n_b - B(-\Delta)n_a]$, $B(x) = x/(e^x - 1)$; ohmic Dirichlet $n = \frac{1}{2}(N_{\text{net}} + \sqrt{N_{\text{net}}^2 + 4n_i^2})$, $p = n_i^2/n$; Neumann at insulating surfaces. **Mobility:** Caughey-Thomas low field $\mu = \mu_{\text{min}} + (\mu_{\text{max}} - \mu_{\text{min}})/(1 + (N/N_{\text{ref}})^a)$ (electrons $68.5/1414/9.2 \times 10^{16}/0.711$;

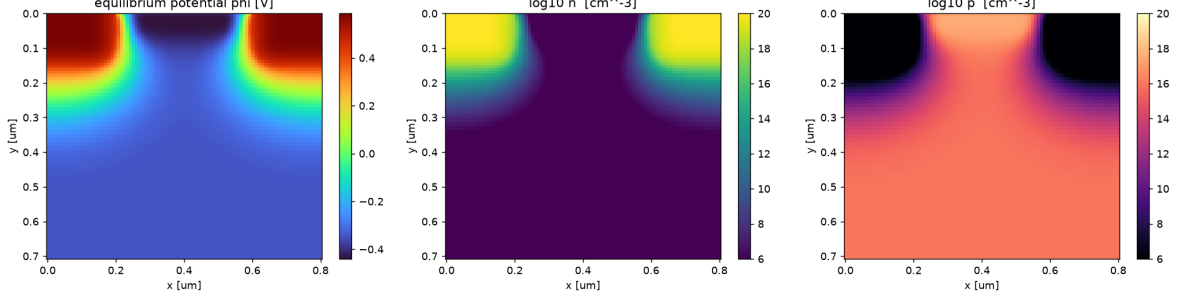


Figure 3: Equilibrium Poisson: potential, $\log_{10} n$, $\log_{10} p$.

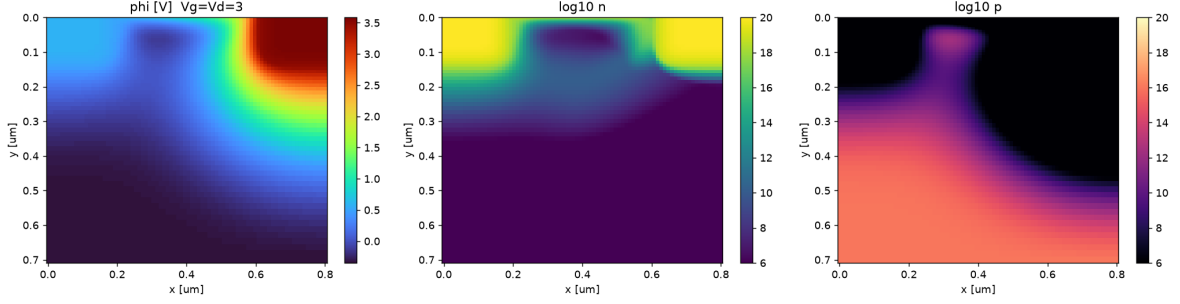


Figure 4: Self-consistent DD at $V_{gs} = V_{ds} = 3\text{ V}$: inversion layer and drain-side high-field / pinch-off region.

holes $44.9/470.5/2.23 \times 10^{17}/0.719$), field saturation $\mu(E) = \mu_{lf}/(1 + (\mu_{lf}E/v_{\text{sat}})^\beta)^{1/\beta}$ ($v_{\text{sat}}^n = 1.07 \times 10^7\text{ cm/s}$, $\beta_n = 2$; $v_{\text{sat}}^p = 8.37 \times 10^6\text{ cm/s}$, $\beta_p = 1$). **Recombination:** SRH, $R = (np - n_i^2)/[\tau_p(n + n_i) + \tau_n(p + n_i)]$, $\tau_n = \tau_p = 10^{-7}\text{ s}$, semi-implicit. **Gummel:** decoupled Poisson $\rightarrow n \rightarrow p$ map, tolerance 10^{-5} V . **Contact-current sign:** drain electron current = $\sum$ SG electron fluxes entering drain-contact nodes, reported per μm width ($\times 10^{-4}$ from A/cm). **SHE (Part II):** Eq. (7) in $(\mathbf{r}, H)$; Maxwellian injection at ohmic contacts, zero-flux turning points at $\varepsilon = 0$, $f_0(\varepsilon_{\text{max}}) \rightarrow 0$.

B Code and parameter provenance

[Rev.2 fix: F8] Modules (Python 3.13, NumPy/SciPy): `constants.py`, `device.py`, `poisson.py`, `dd.py`, `bands.py`, `scattering.py`, `she_bulk.py`, `diagnostics.py`. Each figure is regenerated by running the corresponding module; benchmark numbers in Table 3 are printed by `bands.py` and `diagnostics.py`. Reproduction study, Revision 2.

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